

SAN BERNARDINO, CA, USA

WMO: 722866

Lat: 34.095N

Lon: 117.235W

Elev: 1159

StdP: 14.09

Time Zone: -8.00 (NAP)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	34.5	37.0	5.6	7.7	54.0	9.5	9.3	58.4	27.1	55.7	23.4	58.4	1.4	60	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.0	103.5	68.7	100.4	67.8	98.3	67.4	73.3	95.1	71.5	93.3	70.1	91.8	8.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	99.8	84.5	64.1	93.4	81.5	62.7	88.9	79.3	37.4	95.6	36.0	93.0	34.6	91.7	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.5	13.8	11.7	DB	29.8	109.7	3.2	4.2	27.5	112.7	25.6	115.1	23.8	117.5	21.4	120.5	(4)
(5)			WB	24.4	75.8	2.1	2.1	22.8	77.3	21.6	78.5	20.4	79.7	18.9	81.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.7	54.1	55.3	58.9	65.5	68.1	74.9	80.8	80.4	76.5	70.0	61.4	53.9	(6)	
	DBStd	11.22	5.33	6.46	5.92	5.71	5.84	5.44	5.54	5.16	6.25	5.94	6.15	5.78	(7)	
	HDD50	76	19	22	4	0	0	0	0	0	0	0	2	29	(8)	
	HDD65	1422	338	276	204	62	34	1	0	0	3	19	141	344	(9)	
	CDD50	6178	147	170	280	465	562	747	954	943	795	621	343	151	(10)	
	CDD65	2049	1	3	16	77	131	298	489	478	348	175	33	0	(11)	
	CDH74	26034	90	185	418	1264	1628	3530	5991	5746	4159	2240	680	103	(12)	
(13)	CDH80	14331	9	33	133	550	760	1954	3625	3446	2419	1154	235	13	(13)	
(14)	Wind	WSAvg	3.3	2.8	3.0	3.0	3.4	3.8	4.0	3.8	3.3	2.8	2.8	2.9	(14)	
(15) Precipitation	PrecAvg	14.8	2.9	2.8	2.8	1.2	0.4	0.1	0.0	0.0	0.3	0.6	1.9	2.2	(15)	
	PrecMax	32.0	13.6	12.5	9.6	5.4	2.7	0.5	0.4	0.4	1.3	3.5	8.5	9.1	(16)	
	PrecMin	6.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	6.9	3.0	3.0	2.2	1.3	0.6	0.1	0.1	0.1	0.4	0.9	1.8	2.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.4	84.4	90.4	96.4	99.0	104.6	108.5	107.4	105.3	101.0	91.1	81.8	(19)	
		MCWB	54.5	57.6	60.0	61.9	64.0	67.8	69.6	69.9	67.1	63.9	58.8	53.0	(20)	
	2%	DB	76.4	79.7	83.7	91.1	93.4	99.7	102.5	102.2	101.3	96.0	86.3	76.9	(21)	
		MCWB	52.8	55.0	58.7	60.9	64.0	66.7	69.9	68.9	67.6	62.1	57.2	51.8	(22)	
	5%	DB	72.4	75.2	78.8	86.3	88.8	95.5	99.8	99.2	98.0	91.5	81.7	72.4	(23)	
		MCWB	51.5	53.5	56.7	60.0	63.3	66.8	68.8	68.6	66.5	60.8	55.3	50.3	(24)	
	10%	DB	67.7	70.2	73.4	81.6	83.2	91.2	96.3	95.5	93.4	86.5	77.0	67.7	(25)	
		MCWB	49.9	52.2	54.8	57.6	62.1	66.1	68.6	68.2	65.9	60.5	53.6	49.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.1	61.9	63.0	65.8	68.9	72.1	75.4	75.8	72.6	67.9	62.5	57.3	(27)	
		MCDB	67.3	77.9	78.9	85.5	88.8	94.8	99.4	95.6	95.8	86.6	80.2	65.5	(28)	
	2%	WB	56.2	57.9	60.6	63.5	66.4	70.3	73.2	73.6	70.6	66.1	59.6	55.0	(29)	
		MCDB	65.9	68.4	76.9	84.9	86.8	93.5	95.0	95.1	94.0	84.4	75.3	65.5	(30)	
	5%	WB	54.3	55.8	59.1	61.8	64.7	68.4	71.3	71.8	69.0	64.6	58.1	53.3	(31)	
		MCDB	65.5	67.6	73.9	81.0	84.6	91.4	92.5	93.3	91.8	82.5	73.2	64.6	(32)	
	10%	WB	52.6	54.1	57.2	60.1	63.0	66.6	69.9	69.9	67.4	63.1	56.6	51.8	(33)	
		MCDB	64.4	66.6	68.9	78.0	81.2	88.3	91.0	91.1	88.4	79.4	71.2	63.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	24.2	22.7	22.9	26.7	25.6	30.3	31.0	31.2	30.1	28.5	27.9	24.7	(35)	
		MCDBR	33.4	33.9	34.9	37.6	36.8	36.4	33.4	33.3	36.9	38.3	36.1	34.2	(36)	
		MCWBR	16.3	16.2	16.1	15.8	14.8	13.0	10.4	10.1	13.2	15.5	16.3	17.3	(37)	
	5% WB	MCDBR	23.5	25.2	27.9	31.4	31.8	34.0	30.4	30.7	32.4	30.1	27.2	24.6	(38)	
		MCWBR	13.1	13.4	14.2	14.5	13.5	14.0	11.0	11.2	12.0	13.7	13.8	13.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.297	0.307	0.328	0.344	0.357	0.358	0.388	0.371	0.350	0.338	0.304	0.305	(40)	
	taud		2.539	2.523	2.446	2.407	2.401	2.414	2.396	2.430	2.468	2.472	2.556	2.544	(41)	
	Ebn at Noon		292	301	301	298	294	292	283	287	289	285	287	282	(42)	
	Edh at Noon		26	29	34	37	38	37	38	36	33	30	25	24	(43)	
(44) All-Sky Solar Radiation	RadAvg		1033	1324	1784	2177	2407	2587	2438	2299	1974	1516	1151	937	(44)	
	RadStd		71	125	85	87	112	118	115	70	61	84	52	62	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	+1.13	N/A	-3.06	+1.30	N/A	N/A	N/A	-202	+401	+202

Nomenclature: See separate page